Section 6.6 — Function Transformations - Answer Key

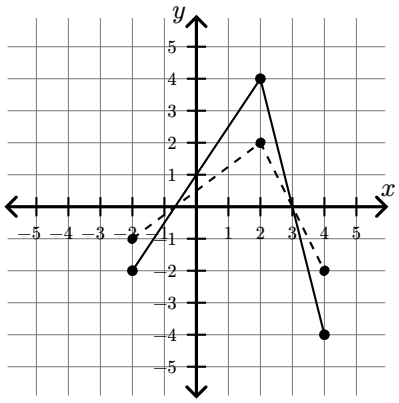
Math 103: College Algebra

Note: In the first two sections, the given graph is shown dashed.

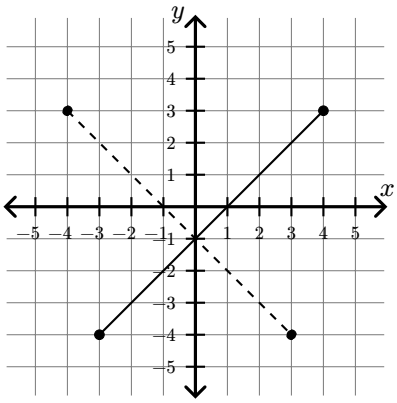
Single-Step Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points.

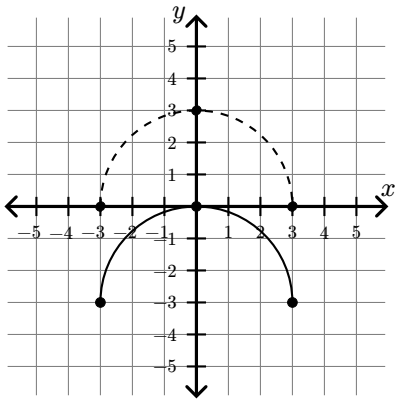
1. $g(x) = 2f(x)$



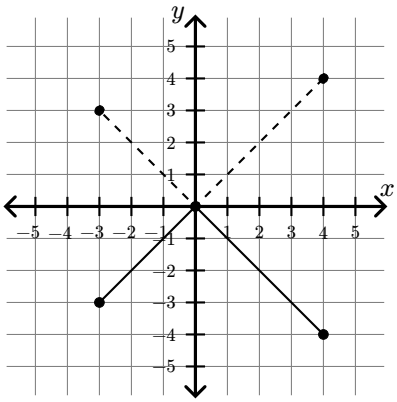
2. $g(x) = f(-x)$



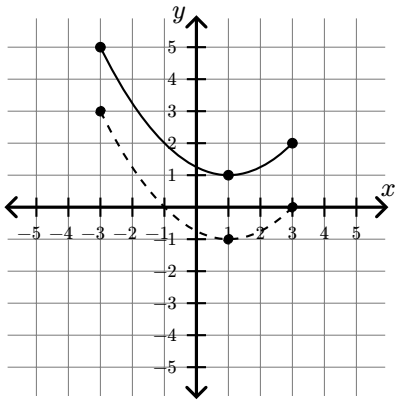
3. $g(x) = f(x) - 3$



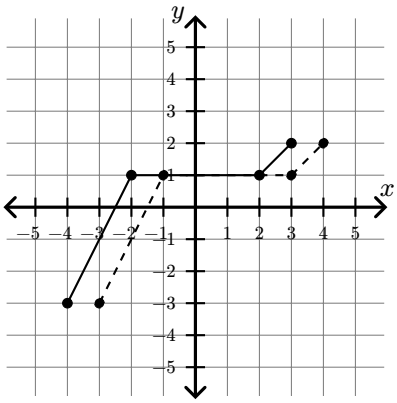
4. $g(x) = -f(x)$



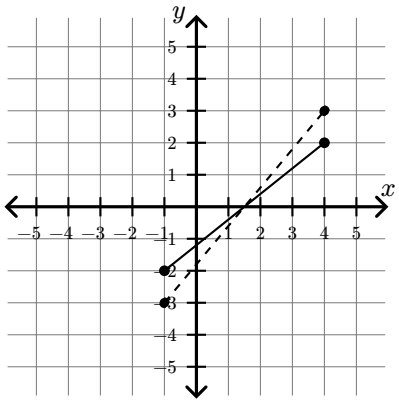
5. $g(x) = f(x) + 2$



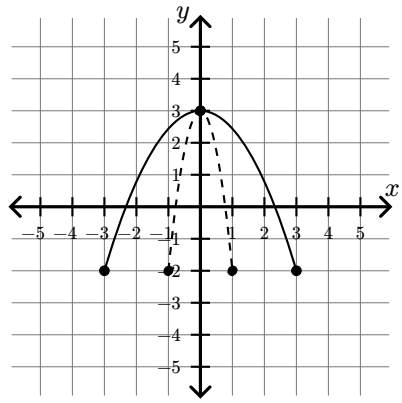
6. $g(x) = f(x + 1)$



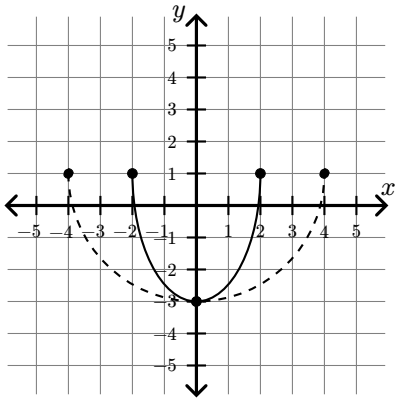
7. $g(x) = \frac{2}{3}f(x)$



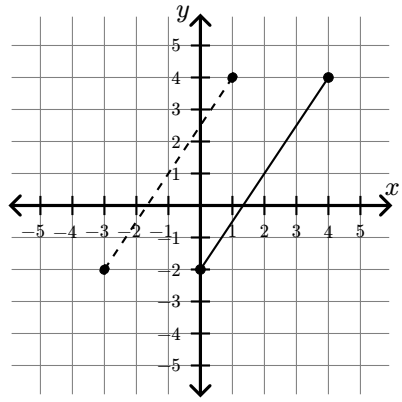
8. $g(x) = f(\frac{1}{3}x)$



9. $g(x) = f(2x)$



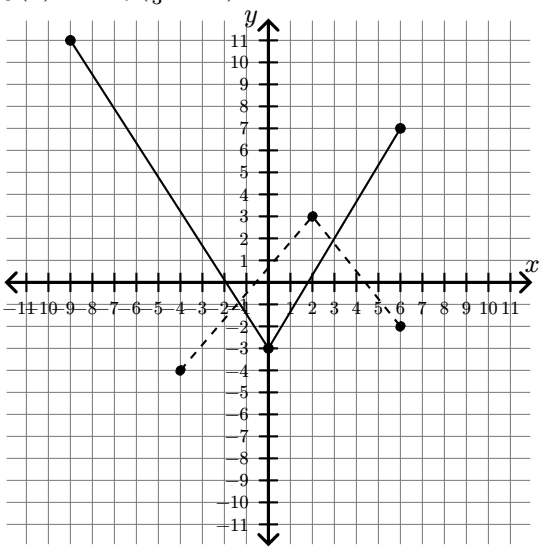
10. $g(x) = f(x-3)$



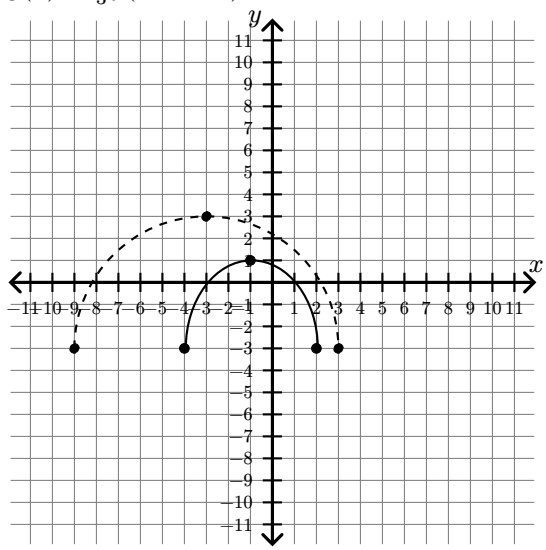
Multiple Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points. *You will want to write out your work on a separate sheet of paper.*

11. $g(x) = -2f(\frac{2}{3}x + 2) + 3$



12. $g(x) = \frac{2}{3}f(-2x - 5) - 1$



Domain and Range

13. $\operatorname{Dom} g = (-\infty, 1)$

14. $\operatorname{Dom} h = (-\infty, -6] \cup \left[-\frac{16}{3}, \infty\right)$

15. $\operatorname{Dom} k = \left(15, \frac{35}{2}\right)$, $\operatorname{Rng} k = (-\infty, \infty)$

Symmetry

Classify each function as even, odd, or neither.

16. Even	17. Neither	18. Odd	19. Odd
20. Even	21. Neither	22. Neither	23. Odd
24. Neither	25. Odd	26. Even	27. Neither

28. A polynomial is an even function if all powers of x are even (including a possible constant). It is odd if all powers of x are odd (and there is no constant). If there is a mix, it is neither.

Using the Library

For each function below:

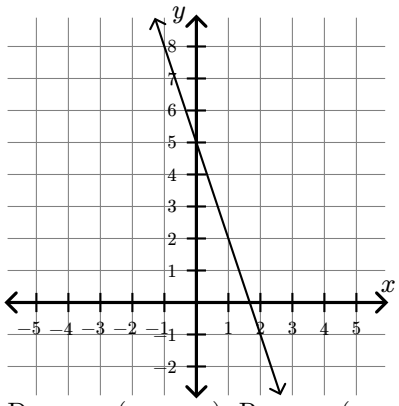
a. Identify the base function (call it $f(x)$) and the family of functions it belongs to.

b. Rewrite the function in terms of $f(x)$. (We usually won't do this explicitly in practice.)

c. Sketch a graph of the function, including any asymptotes and clearly marking any special points, using a separate sheet of graph paper or the other side of this page.

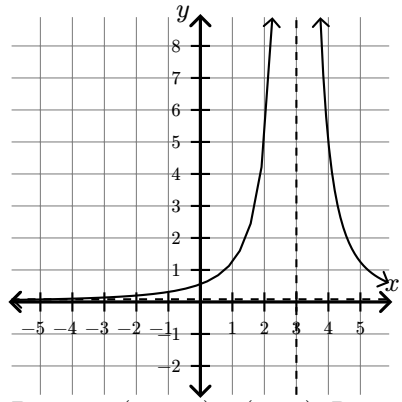
d. Give the domain and range of the function.

29. a. $f(x) = x$ (Linear)
    b. $g(x) = -3f(x) + 5$
    c.



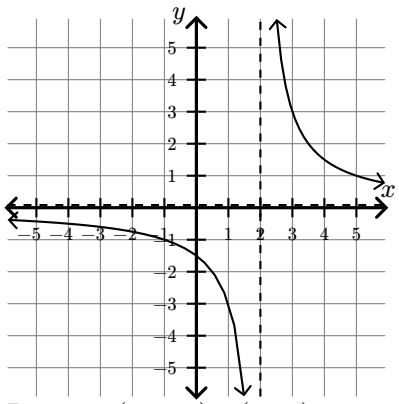
    d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, \infty)$

30. a. $f(x) = \dfrac{1}{x^2}$ (Power (negative even))
    b. $g(x) = \dfrac{5}{(x-3)^2} = 5f(x-3)$
    c.



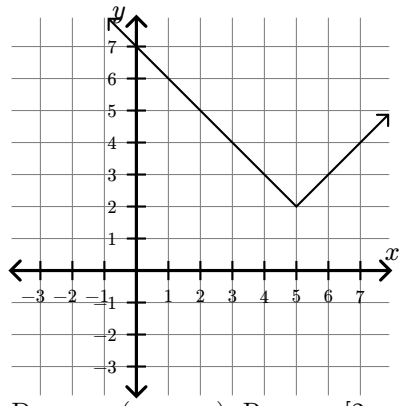
    d. $\operatorname{Dom} g = (-\infty, 3) \cup (3, \infty)$, $\operatorname{Rng} g = (0, \infty)$

31. a. $f(x) = \dfrac{1}{x}$ (Power (negative odd))
b. $g(x) = 3f(x-2)$
c.



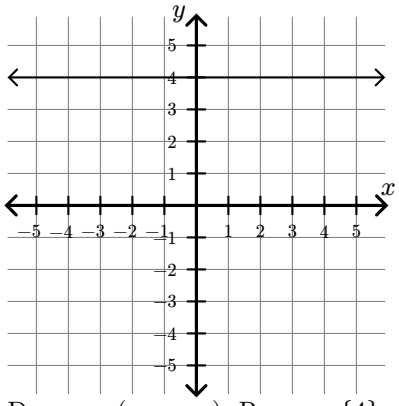
d. $\operatorname{Dom} g = (-\infty, 2) \cup (2, \infty)$,
$\operatorname{Rng} g = (-\infty, 0) \cup (0, \infty)$

32. a. $f(x) = |x|$ (Absolute value)
b. $g(x) = f(x-5) + 2$
c.



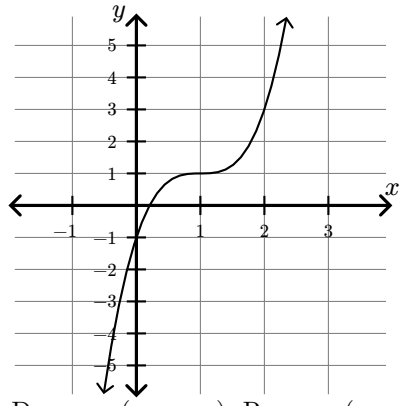
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = [2, \infty)$

33. a. $f(x) = 1$ (Constant)
b. $g(x) = 4f(x)$
c.



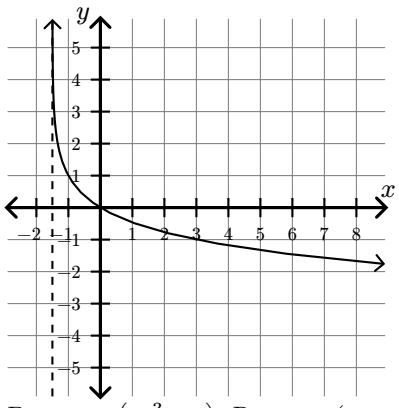
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = \{4\}$

34. a. $f(x) = x^3$ (Power (odd positive))
b. $g(x) = 2f(x-1) + 1$
c.



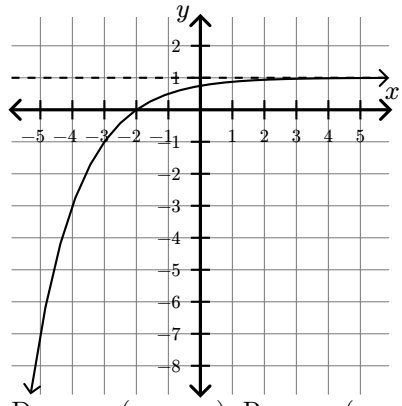
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, \infty)$

35. a. $f(x) = \log_{1/3}(x)$ (Logarithm)
b. $g(x) = f(2x+3) + 1$
c.



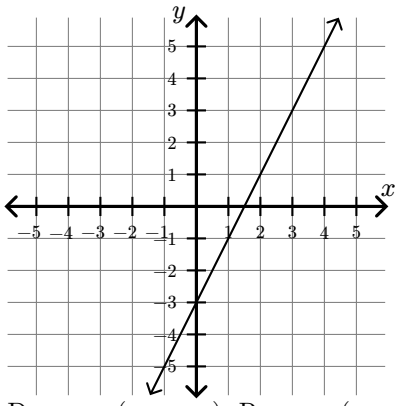
d. $\operatorname{Dom} g = \left(-\frac{3}{2}, \infty\right)$, $\operatorname{Rng} g = (-\infty, \infty)$

36. a. $f(x) = \left(\frac{1}{2}\right)^x$ (Exponential)
b. $g(x) = -f(x+2) + 1$
c.



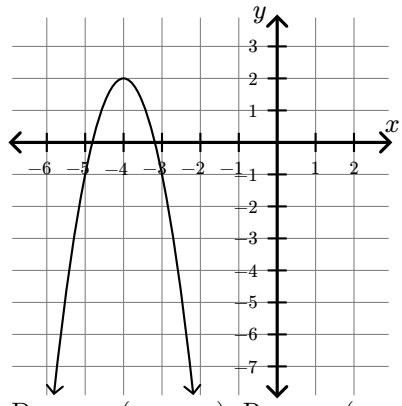
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, 1)$

37. a. $f(x) = x$ (Linear)
b. $g(x) = 2f(x-2)+1$
c.



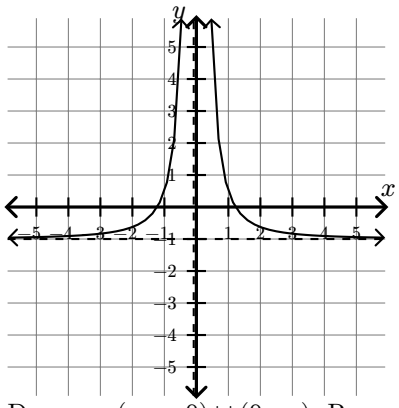
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, \infty)$

38. a. $f(x) = x^2$ (Power (even positive))
b. $g(x) = -3f(x+4)+2$
c.



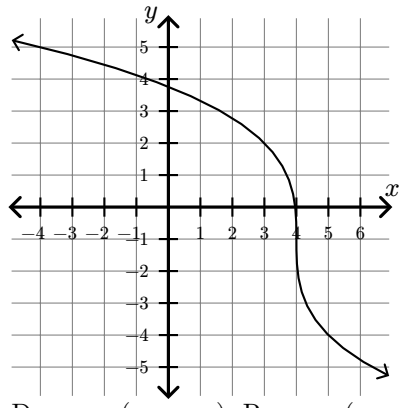
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, 2]$

39. a. $f(x) = \dfrac{1}{x^2}$ (Power (even negative))
b. $g(x) = \frac{3}{2}f(x) - 1$
c.



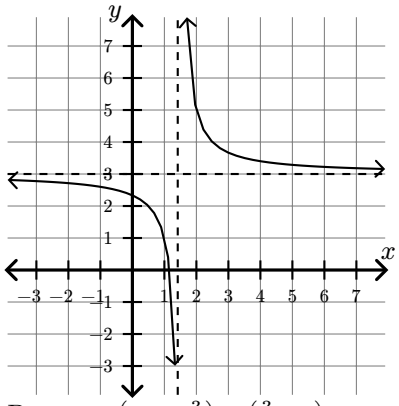
d. $\operatorname{Dom} g = (-\infty, 0) \cup (0, \infty)$, $\operatorname{Rng} g = (-1, \infty)$

40. a. $f(x) = \sqrt[3]{x}$ (Power (odd root))
b. $g(x) = 3f(4-x)-1$
c.



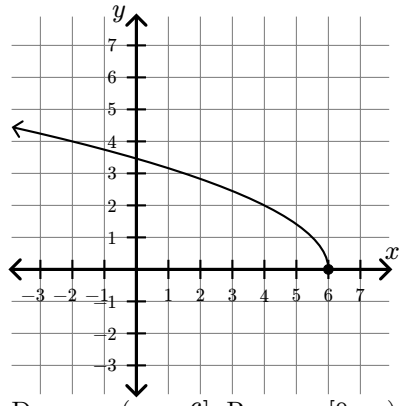
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, \infty)$

41. a. $f(x) = \dfrac{1}{x}$ (Power (odd negative))
b. $g(x) = \dfrac{2}{2x-3} + 3 = 2f(2x-3)+3$
c.



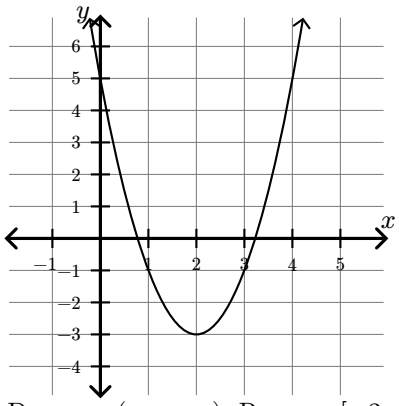
d. $\operatorname{Dom} g = \left(-\infty, \frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right)$,
$\operatorname{Rng} g = (-\infty, 3) \cup (3, \infty)$

42. a. $f(x) = \sqrt{x}$ (Power (even root))
b. $g(x) = 2f(-\frac{1}{2}x+3)$
c.



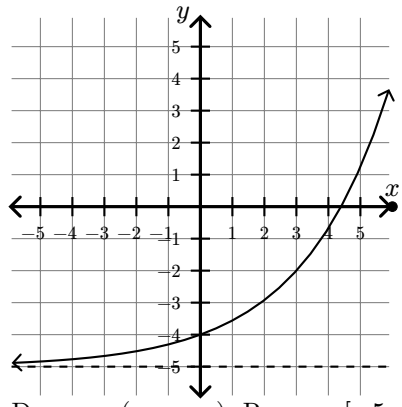
d. $\operatorname{Dom} g = (-\infty, 6]$, $\operatorname{Rng} g = [0, \infty)$

43. a. $f(x) = x^2$ (Power (even positive))
b. $g(x) = 2(x-2)^2 - 3 = 2f(x-2) - 3$
c.



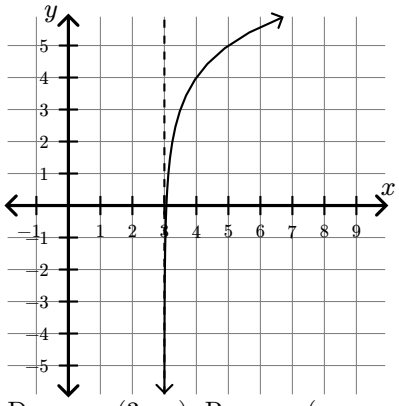
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = [-3, \infty)$

44. a. $f(x) = 3^x$ (Exponential)
b. $g(x) = f(\frac{1}{3}x) - 5$
c.



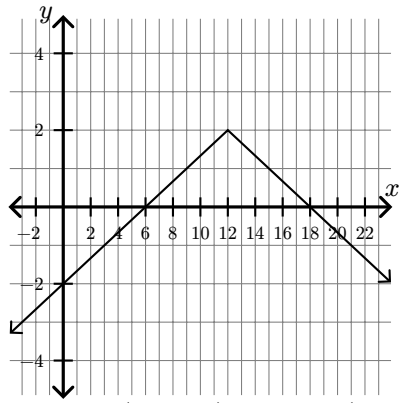
d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = [-5, \infty)$

45. a. $f(x) = \log_2(x)$ (Logarithm)
b. $g(x) = f(x-3) + 4$
c.



d. $\operatorname{Dom} g = (3, \infty)$, $\operatorname{Rng} g = (-\infty, \infty)$

46. a. $f(x) = |x|$ (Absolute value)
b. $g(x) = -f(4 - \frac{1}{3}x) + 2$
c.



d. $\operatorname{Dom} g = (-\infty, \infty)$, $\operatorname{Rng} g = (-\infty, 2]$